

Introduction To Project

1.1 Introduction

We have designed and developed application for Mazi Sheti – Farm Schedule System. It is a web-based application to address today's digital age farming needs. The application –is the initial phase designed to introduce farmers to track their farming activities digitally. This will allow the farmers to maintain their data which can be used later to better manage the farming activities for maximum yield as also it will serve as basis for data analysis for improving productivity, managing costs and reducing the damage due to natural calamities.

The goal of the system is to assist the farmers in organizing their daily, weekly and monthly farm activities. By using our system, farmer can easily track key activities like planting, irrigation, fertilizing, harvesting and completed this task on time. With the help of this system farm operations can be streamlined, leading to optimized time management and better resource utilization.

The system will also allow farmer to schedule, track and receive reminders for important farming activities. Our system provides detail crop information. With the help of this detail descriptions of each crop, farmer can plan their activities effectively. Farmer can also add schedule for a specific crop and plan their farm activities with the help of schedule table.

To use the system, there are two main forms which is login form and registration form. Unless you have an account, you can't login hence you are required to register first. New farmers will need admin approval before they can log in and access the system.

This system makes farm management easier and more organized, helping farmers plan better and increase productivity.

1.2 Existing System

- ✓ Farmers don't always get real-time information about crop health or farm condition, making it harder to make quick decision.
- ✓ These processes can be inefficient, leading to delays in planting, watering and harvesting, which affects overall farm productivity.
- ✓ Farmer often manually record planting and harvesting dates, making it difficult to stay organized.
- ✓ It's hard for farmers to detect and manage crop diseases early, which can cause bigger losses.
- ✓ Traditional methods may not help farmers use less water or reduce chemical use for a more sustainable farm.
- ✓ Without real-time information, it can be difficult for farmers to adjust their schedules quickly in case of changes.

1.3 Need and Scope of the system

- ✓ The system organizes farming tasks like planting, irrigation, and harvesting for better time management.
- ✓ Reduce confusion and provide clear schedules and reminders for important farm activities.
- ✓ Providing detail information about the crop helps farmers to plan better and manage resources effectively.
- ✓ Simplify the registration process for new farmers with admin approval for secure access to the system.
- ✓ Increase productivity by making sure task gets done on time, helping farmers avoid mistakes and delays.

Proposed System

2.1 Objectives

The main object of the Application for Mazi Sheti - Farm Schedule System is to help farmers manage their farm task in an easy and organized way. The system allows farmers to plan and schedule important activities like planting, watering, fertilizing, and harvesting, making sure they complete tasks on time. By using this system farmers giving a clear plan and remainders, which helps them avoid confusion and delays and improving the overall productivity of their farm. The system provides detailed information about each crop, so farmers can make better decisions based on what their crops need. The main goal of the system is to enhance farm management, improve resource utilizations, improve productivity, leading to better yields and greater profits. The system is an easy-to-use that helps farmers manage their tasks better, increase productivity, and improve their farming success.

- **Features of Application for Mazi-Sheti system:**

1. Provide real time information accurate information to the farmers regarding the crops they can choose, crop-based information about nutrients, soil, schedule and climate, seed type etc.
2. Provides the farm size-based recommendations for seeds and seed-types.
3. Reminder emails for the farmer to track schedule and activities.
4. Excel sheet based schedule upload facility for the admin.
5. Real time Reports.

2.2 Requirement Engineering:

Requirement Engineering is the process of finding out what the farmers need from the **Application for Mazi Sheti – Farm Schedule System**, writing these needs down clearly, and making sure the system will work as expected. It helps us understand the farmers' problems, check if the system can be built, decide the best way to solve these problems, and keep track of all the details as the system is created. In simple terms, Requirement Engineering makes sure we know exactly what the system should do, how it should behave, and what limits it has, so that the final system helps farmers manage their farm activities easily and on time.

Feasibility Study: The objective behind the feasibility study is to justify the development of the Mazi Sheti – Farm Schedule System by ensuring the software will be useful and acceptable to farmers, flexible enough to accommodate future changes, and designed following good standards.

- **Technical Feasibility** - Technical feasibility evaluates whether the current technologies are capable of meeting the needs of farmers by supporting features like activity scheduling, reminders, and crop information within the available time and budget.
- **Operational Feasibility** - Operational feasibility assesses how well the system will perform in helping farmers organize their daily, weekly, and monthly farm activities to improve productivity and resource use, making farm management easier and more efficient.
- **Economic Feasibility** - Economic feasibility decides whether the system will be financially beneficial by helping farmers increase crop yields, reduce losses, and manage costs, thereby providing value that justifies the development and maintenance expenses.

2.2.1 Requirement Gathering

Requirement gathering is the process of collecting all the important information and needs from the farmers who will use the **Application for Mazi Sheti – Farm Schedule System**. This helps us understand exactly what features and tools the system should have to make farm management easier.

For this system, requirement gathering includes:

1. Talking to farmers to learn about their daily farm activities like planting, watering, fertilizing, and harvesting.
2. Finding out what problems they face in keeping track of these activities.
3. Understanding how they want to schedule and get reminders for their tasks.
4. Collecting information about what crop details farmers want to see in the system.
5. Learning about user needs, like registration and admin approval before accessing the system.
6. Writing down all these needs clearly to build a system that truly helps farmers organize and improve their farm work.

2.2.2 Software Requirements

Frontend:

- React.js – Used to build a dynamic and user-friendly interface for inputting course outlines.
- Tailwind CSS – A utility-first CSS framework to design a responsive, clean, and modern UI.

Backend:

- Spring Boot – Provides the runtime environment to run server-side Java code.
- Spring Web – A lightweight web framework for handling backend logic and RESTful API endpoints.

Database:

- MySQL – A relational database used to store structured data such as courses, units, instructors, and outline details.

Authentication & Security:

Development Tools:

- Visual Studio Code – The primary code editor used for writing and managing the full stack application.
- Postman – Useful for testing APIs during development to ensure proper communication between frontend and backend.

2.2 Software Requirements Specification (SRS):

The Application for Mazi Sheti – Farm Schedule System is a web – based application designed to help farmers manage their daily, weekly, and monthly farm activities in a digital and organized manner. This system assists farmers in tracking their tasks, scheduling activities, and receiving reminders, and improving productivity.

1. Use Management:

- The system includes Admin and Farmer roles.
- Admin can approve new farmers to join the system after registration.
- Farmers can log into the system after getting approval from Admin.

2. Farm Activity Management:

- Farmers can schedule and track tasks like planting, watering and harvesting.
- The system shall provide farmers with a schedule table for better planning of farm activities.
- Each crop shall have a detailed description with farming information to assist in planning and scheduling.

3. Crop Management:

- The system manages crop details including planting season, growth duration, and other relevant attributes.
- The system shall provide farmers with detailed crop information, including optimal growth conditions, irrigation schedules and harvesting guidelines.

4. Reminder System:

- The system shall provide reminders for scheduled activities, such as fertilization, to ensure timely task completion.
- Farmers shall be able to receive reminders through email.

5. Data Storage and Management:

- All farming data including tasks and crop schedules, is stored securely.
- Farmers can view past activities to help plan future farming more effectively.

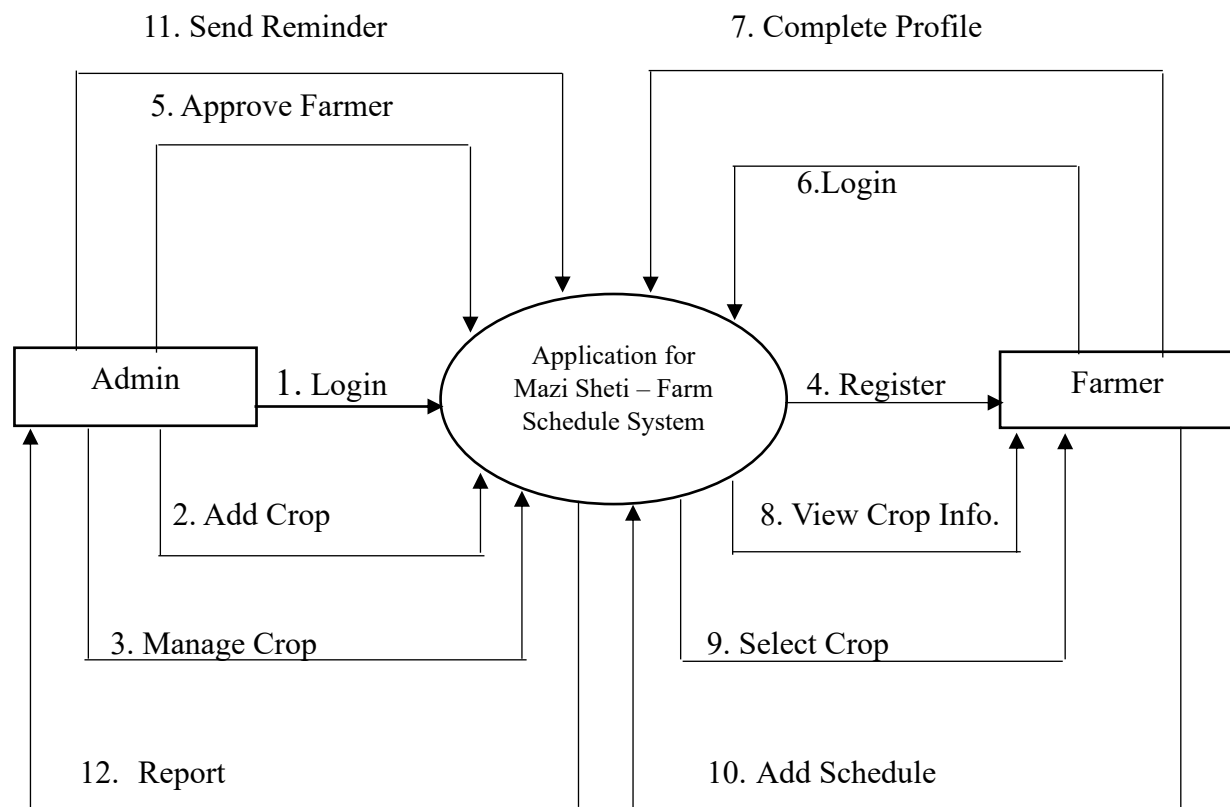
6. Reporting:

- The system generates reports showing crop details, upcoming activities.
- Admin access reports on user activities, while farmers view their own tasks and schedules.

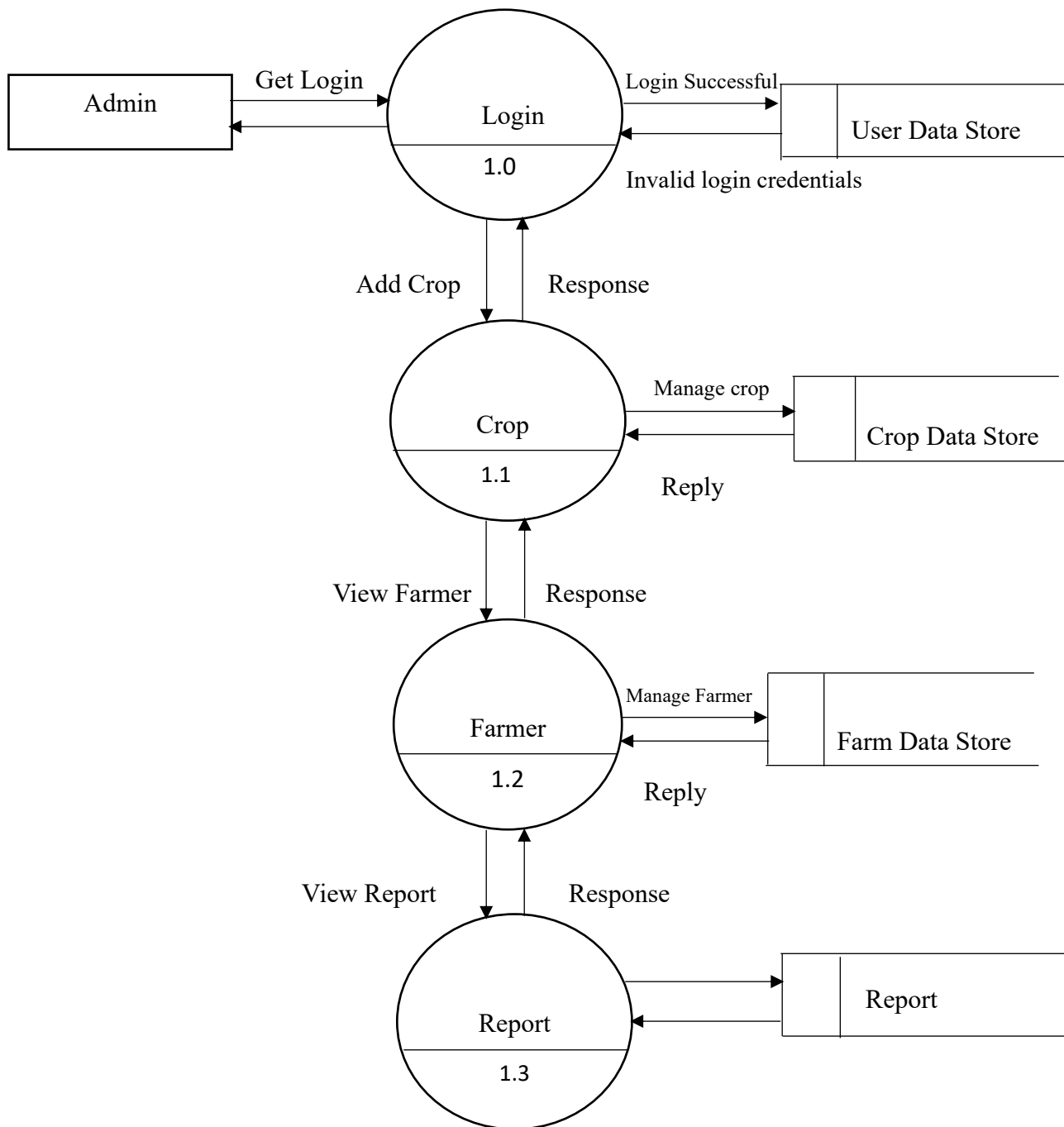
3.1 SYSTEM DIAGRAMS

3.1.1 Data Flow Diagram (DFD)

Context level diagram

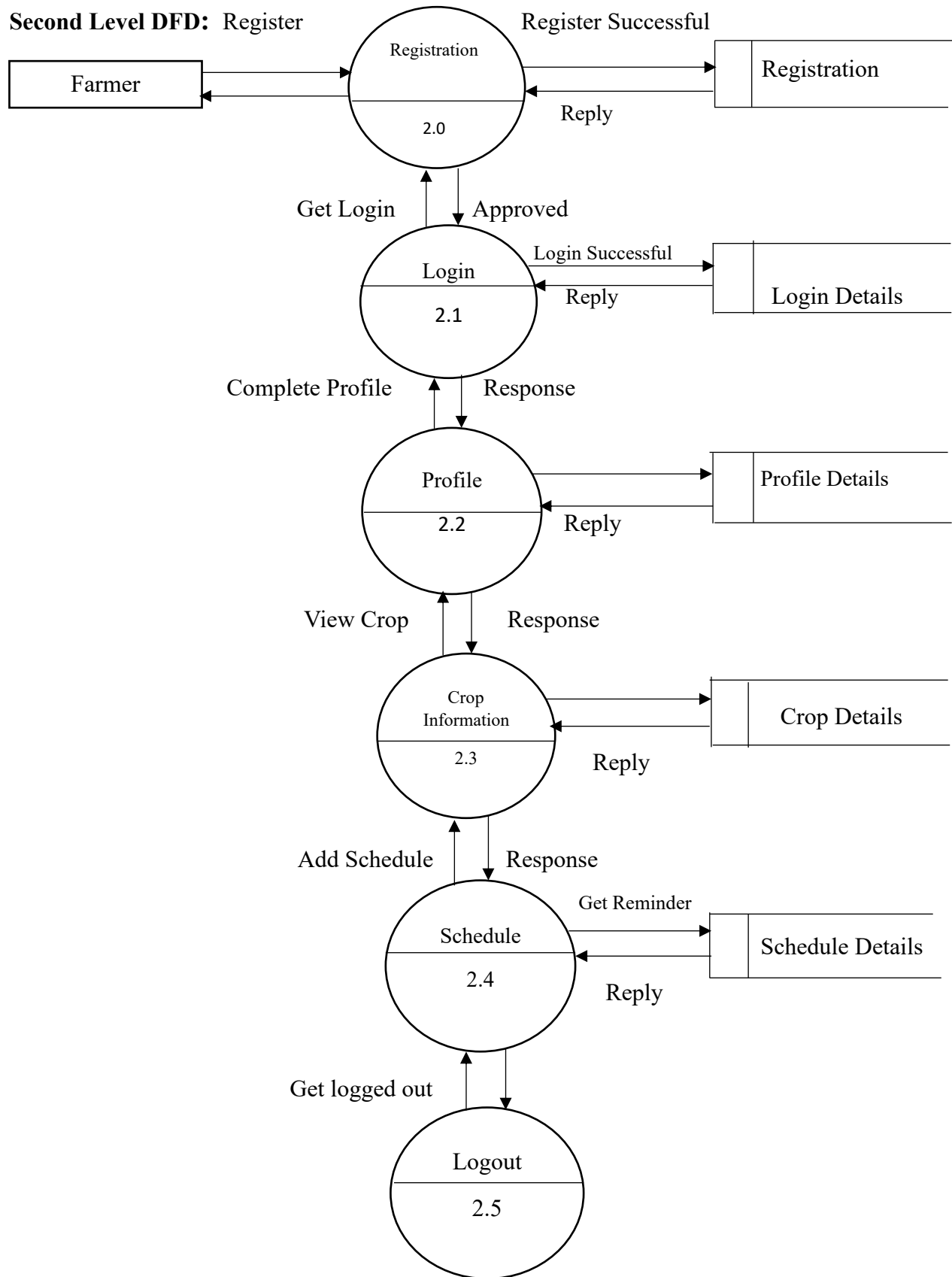


First Level DFD:

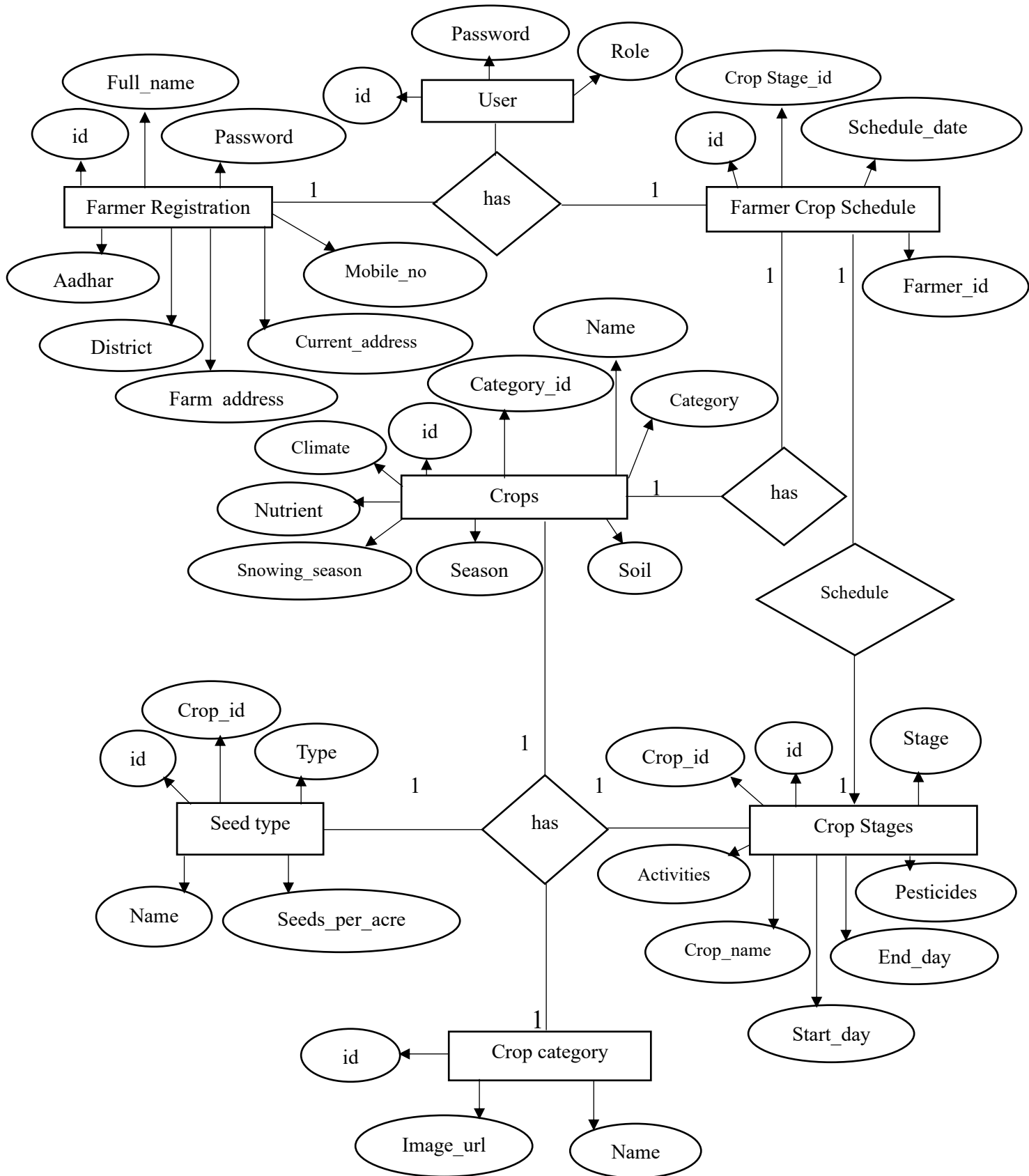


Application for Mazi Sheti- Farm Schedule System

Second Level DFD: Register



3.1.2 ERD (Entity Relationship Diagram):



System Design

4.1 Database Design

Farmer Registration:

System Name: Application for Mazi Sheti – Farm Schedule System				
Table Name: Farmer Registration				
Sr. No	Field Name	Data type	Size	Description
1	Id	int	-	Primary key To store farmer id.
2	Full_name	Varchar	50	Primary key To store the full name
3	Mobile_number	Varchar	100	To store the mobile number
4	Password	Varchar	225	Primary key To store the password
5	Status	Varchar	100	To store the status
6	Current_address	varchar	225	To store the current address
7	Farm_address	varchar	225	To store the farm address
8	district	varchar	225	To store the district name
9	aadhar	varchar	225	To store the aadhar
10	Land_size	double	225	To store the land size
Reports -: Farmer Registration				

User:

System Name: Application for Mazi Sheti – Farm Schedule System				
Table Name: User				
Sr. No	Field Name	Data type	Size	Description
1	Id	int	-	To store user id.
2	Approved	Varchar	50	To store the approved user
3	Email	Varchar	100	To store the email
4	Full_name	Varchar	225	Foreign key from farmer registration table To store the full name
5	Password	int	-	Foreign key from farmer registration table To store password
6	Role	varchar	225	To store the user role

Crop categories:

System Name: Application for Mazi Sheti – Farm Schedule System				
Table Name: Crop categories				
Sr. No	Field Name	Data type	Size	Description
1	Id	int	-	Primary key To store category id
2	Image_url	Varchar	225	To store the images
3	name	Varchar	100	To store the crop name

Crops:

System Name: Application for Mazi Sheti – Farm Schedule System				
Table Name: Crops				
Sr. No	Field Name	Data type	Size	Description
1	Id	int	-	Primary key To store crop id
2	Climate	Varchar	225	To store the images
3	name	Varchar	100	Primary key To store the crop name
4	nutrients	Varchar	225	To store the nutrients
5	Season	Varchar	225	To store the season
6	Snowing_season	Varchar	225	To store the snowing season
7	soil	Varchar	225	To store the soil
8	category	Varchar	225	To store the crop category
9	Image_url	Varchar	225	To store images
10	Category_id	int	-	Foreign key from crop categories table To store category id
Reports: - Crop				

Seed Type:

System Name: Application for Mazi Sheti – Farm Schedule System				
Table Name: Seed Type				
Sr. No	Field Name	Data type	Size	Description
1	Id	int	-	To store user seed id.
2	name	Varchar	255	To store the seed name
3	Seeds_per_acre	Varchar	255	To store the seed per acre
4	Type	Varchar	225	To store the seed type
5	Crop_id	int	-	Foreign key from crops table To store crop id
Report: - Seed Type				

Crop Stages:

System Name: Application for Mazi Sheti – Farm Schedule System				
Table Name: Crop Stages				
Sr. No	Field Name	Data type	Size	Description
1	Id	int	-	Primary key To store crop stage id
2	Activities	Varchar	225	To store the activities
3	Crop_name	Varchar	225	Foreign key from crops table To store the crop name
4	End_day	Varchar	225	To store the end day
5	Pesticides	Varchar	225	To store the pesticides
6	Stage	Varchar	225	To store the stages
7	Start_day	Varchar	225	To store the start day
8	Crop_id	Int	225	Foreign key from crops table To store the crop id
Report: - Crop Stages				

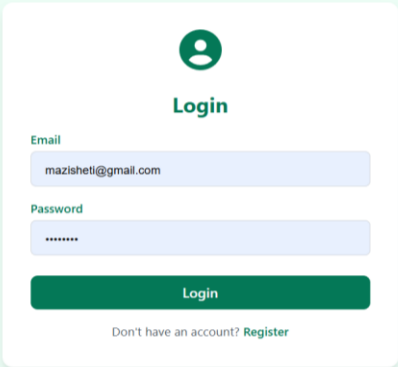
Farmer crop schedule:

System Name: Application for Mazi Sheti – Farm Schedule System				
Table Name: Farmer crop schedule				
Sr. No	Field Name	Data type	Size	Description
1	Id	int	-	To store farmer crop schedule id
2	Schedule_date	Varchar	225	To store the schedule date
3	Crop_stage_id	int	-	Foreign key from crops table To store the crop name
4	Farmer_id	Varchar	225	Foreign key from farmer registration table To store farmer id
Report: - Farmer Crop Schedule				

4.2 Input Design & its samples

Admin Login:

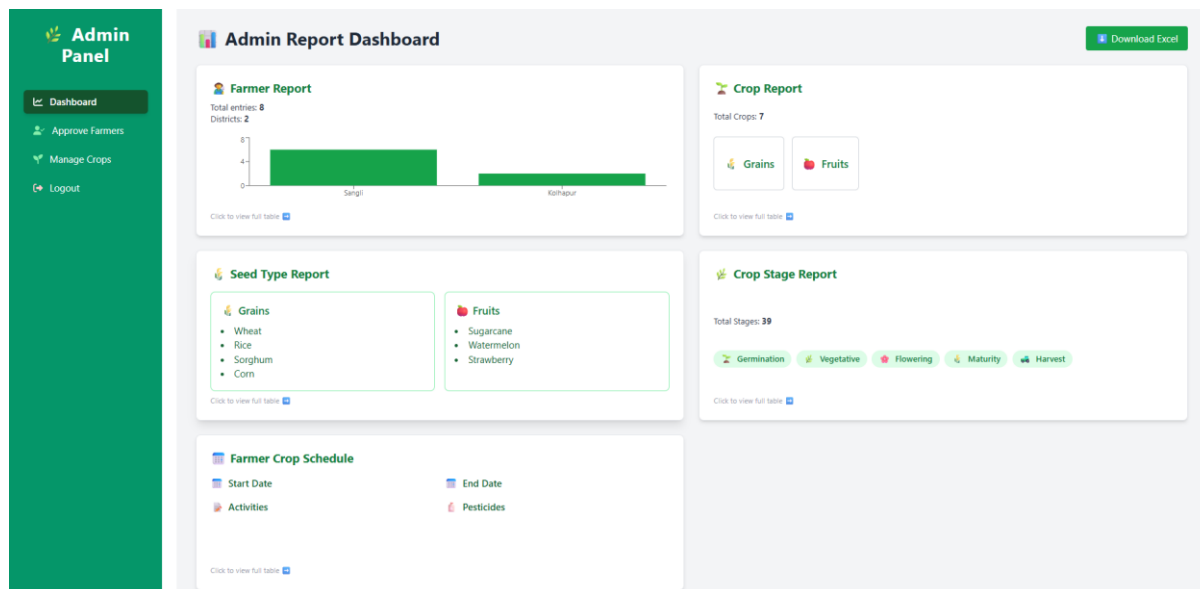
Description: In this page admin can login the application through his credentials



The login form is centered on a light green background. It features a green circular icon with a white person silhouette at the top. Below the icon is the word "Login" in bold green text. The form includes two input fields: "Email" with the value "mazisheti@gmail.com" and "Password" with masked characters "*****". A green "Login" button is positioned below the password field. At the bottom, there is a link that says "Don't have an account? Register".

Admin Dashboard:

Description: In this page, Admin can view the reports.



Approve Farmers:

Description: Admin can approve the new registered farmers.

The screenshot shows the 'Admin Panel' on the left with a green sidebar containing 'Dashboard', 'Approve Farmers', 'Manage Crops', and 'Logout'. The main content area is titled 'Farmer Approval Panel' and features a search bar 'Search by name or email'. Below the search bar is a list of six farmers, each with their name, email, status, and an 'Approved' button.

Name	Email	Status	Action
Snehal Thorat	ravi@example.com	APPROVED	Approved
Rahul Patil	Rahulp66@gmail.com	APPROVED	Approved
Dipak Dadaso Thorat	dipakt088@gmail.com	APPROVED	Approved
Dipak Dadaso Thorat	dipakt088@gmail.com	APPROVED	Approved
Sanika Patil	Sanikap66@gmail.com	APPROVED	Approved
Raj Pawar	rajp06@gmail.com	APPROVED	Approved

Manage Crop:

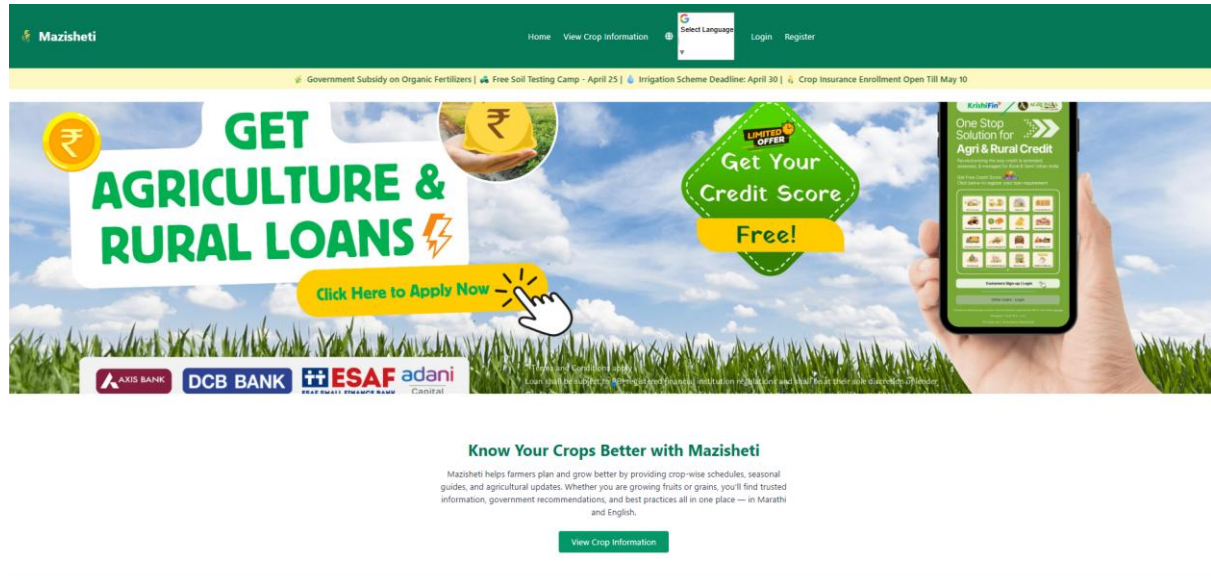
Description: Admin can add crop, upload seed type sheet, upload crop stages sheet.

The screenshot shows the 'Admin Panel' on the left with a green sidebar containing 'Dashboard', 'Approve Farmers', 'Manage Crops', and 'Logout'. The main content area is titled 'Crop & Seed Management' and features three sections: 'Add New Crop', 'Upload Seed Types (Excel)', and 'Upload Crop Stages (Excel)'. The 'Add New Crop' section has a form with fields for 'Grains', 'Corn', 'Loamy or sandy loam soils', 'Nitrogen, phosphorus, and potassium', '21°C to 30°C', 'Kharif season', 'No snowing season', and a file upload field. The 'Upload Seed Types (Excel)' section has a file upload field and an 'Upload Seed Types' button. The 'Upload Crop Stages (Excel)' section has a dropdown menu for 'Corn', a file upload field, and an 'Upload Crop Stages' button.

4.3 Output Design

Homepage:

Description: When admin or farmer login to our website then they will see all the information of homepage.

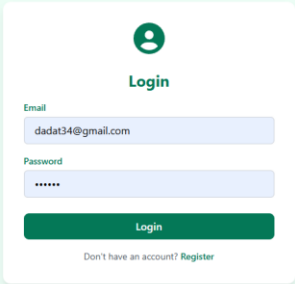


Farmer Registration:

Description: Farmer can register with his details like full name, mobile number, email, password, etc.

Farmer Login:

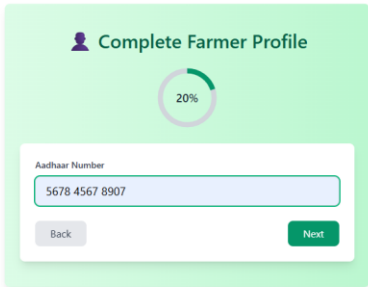
Description: In this page Farmer login with his credentials



The login form is centered on a light green background. It features a green circular icon with a white person silhouette at the top. Below the icon is the word "Login" in green. The form has two input fields: "Email" with the text "dadat34@gmail.com" and "Password" with masked characters "*****". A green "Login" button is positioned below the password field. At the bottom, there is a link that says "Don't have an account? Register".

Farmer Complete Profile:

Description: In this page farmer can complete his profile by adding the details.



The "Complete Farmer Profile" form is displayed on a green background. At the top, it shows a profile icon and the title "Complete Farmer Profile". A circular progress indicator shows "20%". Below this is an input field for "Aadhaar Number" containing "5678 4567 8907". "Back" and "Next" buttons are at the bottom of the form. The page header includes the "Mazisheti" logo, navigation links ("Home", "View Crop Information", "View Saved Schedule"), a language selector, and a "Logout" button. A yellow banner below the header contains several announcements. The footer includes social media links, contact information, and a copyright notice.

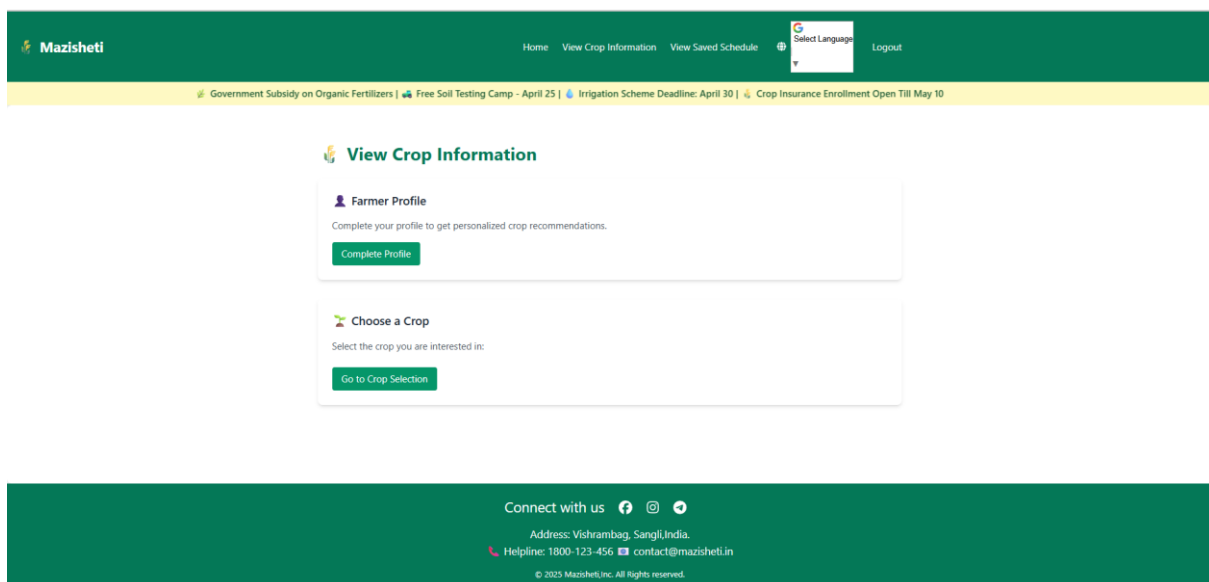
Select Language:

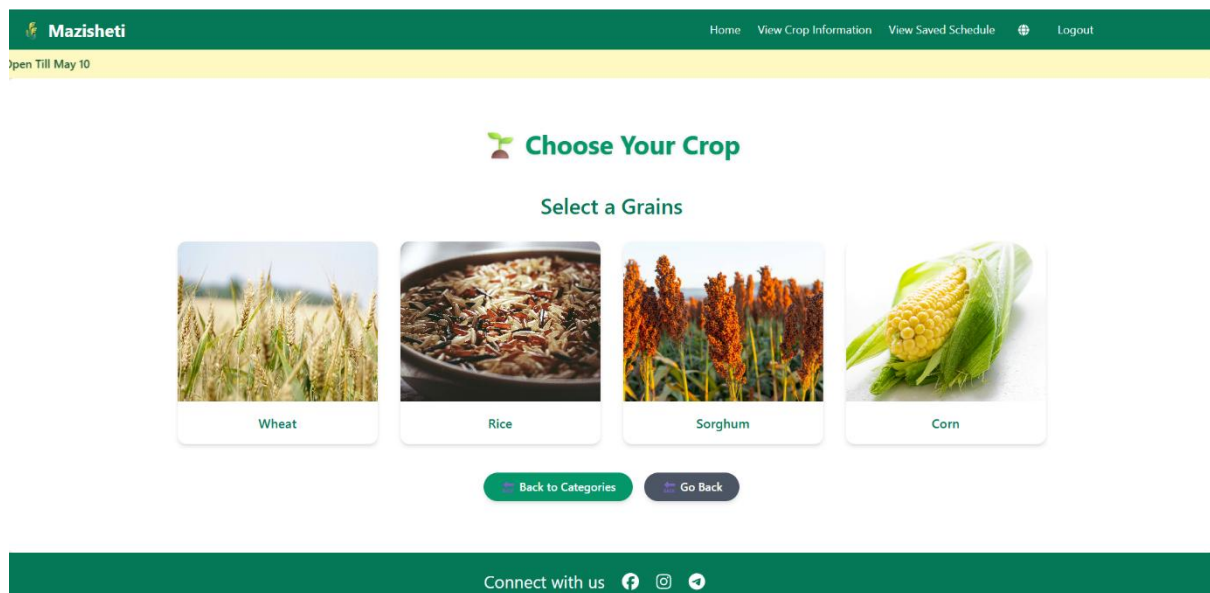
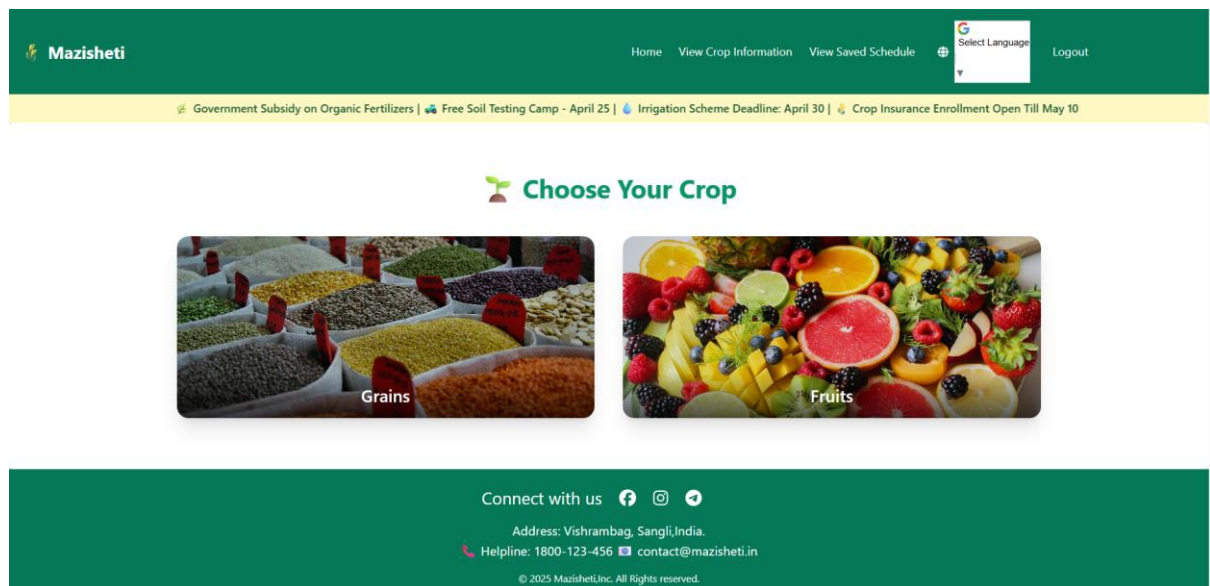
Description: Farmer can choose language.



View crop information:

Description: Farmer can view information of crop in this page.





Mazisheti

HomeView Crop InformationView Saved Schedule

Select Language

Logout

Government Subsidy on Organic Fertilizers

Rice - Crop Details

Soil

Clayey or silty loam soils

Nutrients

nitrogen, phosphorus, and potassium

Season

Kharif crop

Climate

20°C to 37°C.

Snowing Season

Not suitable

Available Seed Types

Indrayani

Type:

Seeds Per Acre: 25 to 30 kilograms per acre

Jaya

Type:

Seeds Per Acre: 25 to 30 kilograms per acre

Sona Masuri

Type:

Seeds Per Acre: 25 to 30 kilograms per acre

Land Size (in acres):

1

You will need approximately 50.00 kilograms of seeds for 1 acre(s).

Proceed to Schedule

Back to Crops

Connect with us

Address: Vishrambag, Sangli, India.

Helpline: 1800-123-456

contact@mazisheti.in

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Application for Mazi Sheti- Farm Schedule System

Farmer Add Schedule:

Description: In this page farmer can add the schedule for selected crop .

Crop Growth Schedule for Rice

Seed Type: Indrayani

Quantity: 50

Select Planting Date: 10-06-2025

☒ Regenerate Schedule

Regenerate Schedule

Stage	Start Date	End Date	Activities	Pesticides / Herbicides
Pre-planting	2025-06-10	2025-06-10	Soil testing, land preparation, and seed selection.	Use pre-emergent herbicides like Pedimethalin to control weeds.
Planting	2025-06-11	2025-06-11	Sow seeds at the recommended depth and spacing.	No pesticides required at this stage.
Germination	2025-06-12	2025-06-20	Ensure adequate moisture for seeds germination. Monitor for pests.	Apply fungicides like Carbendazim if fungal diseases are observed.
Tillering	2025-06-21	2025-07-10	Promote tiller growth with proper irrigation and nutrient management.	Use insecticides like Chlorpyrifos for pest control if needed.
Stem Elongation	2025-07-11	2025-08-09	Apply nitrogen fertilizers to boost growth. Monitor for pests and diseases.	Use herbicides like 2,4-D to manage broadleaf weeds.
Stem Elongation	2025-08-10	2025-09-08	Ensure proper irrigation. Monitor for pests and diseases.	Apply fungicides like Propiconazole for fungal infection.
Heading	2025-09-09	2025-09-28	Ensure even flowering by maintaining optimal conditions.	Use insecticides like Imidacloprid for aphid control if necessary.
Grain Filling	2025-09-29	2025-10-18	Maintain irrigation and nutrient supply. Monitor for pests and diseases.	Apply fungicides like Tebuconazole to prevent fungal diseases.
Ripening	2025-10-19	2025-11-07	Reduce irrigation to allow grains to mature.	No pesticides required at this stage.
Harvesting	2025-11-08	2025-11-17	Harvest the crop when grains are fully mature and moisture content is low.	No pesticides required at this stage.

Download Schedule as PDF

Save Schedule

Farmer Save Schedule:

Description: In this page farmer can save the schedule.

Mazisheti

Home

View Crop Information

View Saved Schedule

Select Language

Logout

Government Subsidy on Organic Fertilizers | Free Soil Testing Camp - April 25 | Irrigation Scheme Deadline: April 30 | Crop Insurance Enrollment Open Till May 10

Saved Crop Schedules

Back to Crop Selection

Stage	Start Date	End Date	Activities	Pesticides
Pre-planting	2025-06-04	2025-06-04	Soil testing, land preparation, and seed selection.	Use pre-emergent herbicides like Pedimethalin to control weeds.
Planting	2025-06-05	2025-06-05	Sow seeds at the recommended depth and spacing.	No pesticides required at this stage.
Germination	2025-06-06	2025-06-14	Ensure adequate moisture for seeds germination. Monitor for pests.	Apply fungicides like Carbendazim if fungal diseases are observed.
Tillering	2025-06-15	2025-07-04	Promote tiller growth with proper irrigation and nutrient management.	Use insecticides like Chlorpyrifos for pest control if needed.
Stem Elongation	2025-07-05	2025-08-03	Apply nitrogen fertilizers to boost growth. Monitor for pests and diseases.	Use herbicides like 2,4-D to manage broadleaf weeds.
Stem Elongation	2025-08-04	2025-09-02	Ensure proper irrigation. Monitor for pests and diseases.	Apply fungicides like Propiconazole for fungal infection.
Heading	2025-09-03	2025-09-22	Ensure even flowering by maintaining optimal conditions.	Use insecticides like Imidacloprid for aphid control if necessary.
Grain Filling	2025-09-23	2025-10-12	Maintain irrigation and nutrient supply. Monitor for pests and diseases.	Apply fungicides like Tebuconazole to prevent fungal diseases.
Ripening	2025-10-13	2025-11-01	Reduce irrigation to allow grains to mature.	No pesticides required at this stage.
Harvesting	2025-11-02	2025-11-11	Harvest the crop when grains are fully mature and moisture content is low.	No pesticides required at this stage.

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5.1 System Requirements

To run the Course Outline Project efficiently, ensure the following system setup:

- A computer or laptop with a stable internet connection.
- Support for Spring boot, MySQL, and a modern browser.
- Ability to run a local or remote backend server (Spring boot).

5.2 Hardware Requirements

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Keyboard: Standard keyboard
- Mouse: Standard quality mouse
- Monitor: Color monitor
- Internet: required for accessing the web application

5.3 Software Requirements

- Operating System: Windows 10/11, Linux (Ubuntu), or macOS
- Web Browser: Google Chrome, Mozilla Firefox, or Microsoft Edge
- Text Editor / IDE: Visual Studio Code

Backend Technologies:

- Spring Boot – Java-based framework for building web application
- Spring MVC – Web Framework for Spring Boot
- MySQL – Relational database for storing courses, teachers, and mappings
- MySQL Workbench / Shell – For managing the MySQL database

Frontend Framework:

- React.js – For building the user interface

Tools:

- npm – Node package manager
- Axios – For HTTP requests
- Postman – For API testing

5.4 Installation Process

Step 1: Install Required Software

Make sure the following software is installed on your system:

- Node.js and npm (Download from <https://nodejs.org>)
- MySQL database (Download from <https://dev.mysql.com/downloads/mysql/>)
- MySQL Workbench (optional but helpful for managing your database)
- Visual Studio Code for editing code (Download from <https://code.visualstudio.com>)

Step 2: Download Project Files

- Get the project folder from your teacher or copy it from a pen drive or Google Drive.
- Save it on your computer, for example in a folder named Mazi Sheti
- Inside this folder, there should be two main folders:
 - frontend (React app)
 - backend (Spring Boot)

Step 2: set up the front-end

- Open a new terminal window.
- Go to the frontend folder: `cd frontend`
- Install all frontend packages: `npm install`
- Create a `.env` file inside the frontend folder with the following content
`VITE_API_URL=http://localhost:5173`
- Start the React development server
- Run this command to start frontend: `npm run dev`

Step 2: set up the backend

- In another terminal window, go to the backend folder: `cd backend`
- Install backend dependencies: `mvnw clean install`
- Create application. Properties file inside the backend folder with the following content

`spring.datasource.url=jdbc:mysql://localhost:3306/mazisheti`

`spring.datasource.username=root`

`spring.datasource.password=Cimdr@12345`

`spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver`

`spring.jpa.hibernate.ddl-auto=update`

`spring.jpa.show-sql=true`

`server.port=8082`

Start the backend server: `spring-boot`

5.5 User Guidelines

- **Admin Login:** Secure login for authorized users.
- **Add Crop:**
 - Enter details like crop name, crop categories, crop stages, crop information, etc.
 - When you click save, the info is sent to the backend and stored in the database.
- **Upload Crop Stages Sheet:**
 - Upload the crop stages sheet
 - Save the stages sheet in the database.
- **Upload the Seed type sheet:**
 - Upload the seed type sheet
 - Save the seed type sheet in the database
- **Approve Farmer:**
 - Approve farmer and send a approve mail

- **Farmer Registration:**
 - Farmer register with details like name, mobile no, Aadhar no, email id, password, etc
- **Farmer Login:**
 - Farmer login with his (login credentials user name and password)
- **Add Schedule:**
 - Add schedule with details like crop name, planting date, etc

Reports

1. Farmer Registration Report:

Description: In this report admin can view the registration details of farmer.

Farmer Report										
Reports Table										
id	fullName	mobileNumber	email	password	status	aadhaar	currentAddress	farmAddress	district	landSize
9	Aaliya shakil betgeri	9613171177	aaliyabetgeri3@gmail.com	Aaliya_07	APPROVED	9997 6778 8034	College corner,Sangli	Bedag,Sangli	Sangli	22
10	Neha Shakil Bagnikar	9860749932	nehabagnikar8@gmail.com	Neha@32	APPROVED	4567 7895 3456	Laxmi market,Miraj	Arag road,Arag	Sangli	10
11	Pratibha Deshmukh	9921350322	pratibha.deshmukh@despune.org	123	APPROVED	8790 2234 5587	M G road,Jat	Mali vasti,Jat	Sangli	20
12	omkar vilas mote	9765755642	omkarvilasmote@gmail.com	ABC@123	APPROVED	7766 8463 2336	Ganshenagar,Kolhapur	Ganganbavda,Kolhapur	Kolhapur	8
21	Divya Dadaso Thorat	09579374987	thoratdivya77@gmail.com	Divya@77	APPROVED	5678 4567 8900	Gundewadi Road ,Malgaon	Ramnager,Malgaon	Sangli	12
22	Arati Sukumar Puneekar	9956788970	aratip67@gmail.com	Puneekar@70	PENDING	6785 6034 2010	Janrawadi,Sangli	Salgar road, Janrawadi	Sangli	4
23	Vishwajit Sanjay Thorat	8890876745	vthorat56@gmail.com	Vishwajit@70	PENDING	7890 4563 4578	Shivaji Chok,Tasgaon	High Road,Shirdon	Sangli	9
24	Jaydip Rajkumar Thorat	09322596614	jaydipthorat140@gmail.com	Jaydip@40	PENDING	2345 5678 9012	Umalwad,Jaysingpur	Akali Road,Jaysingpur	Kolhapur	6

2. Crop Report:

Description: In this report admin can view the crop information details.

Crop Report							
Reports Table							
id	name	soil	nutrients	climate	season	snowingSeason	imageUrl
6	Sugarcane	Sandy loam, clay loam, or even red soils	nitrogen, phosphorus, and potassium	20°C to 30°C	Rabi crop	Mid-November to late December	http://localhost:8082/uploads/images/1747219518798_wheat.jpg
7	Wheat	Loamy or clay loamy soils	nitrogen, phosphorus, and potassium	20°C to 30°C	Rabi crop	Mid-November to late December	http://localhost:8082/uploads/images/1747223761964_wheat.jpg
8	Rice	Clayey or silty loam soils	nitrogen, phosphorus, and potassium	20°C to 37°C	Kharif crop	Not suitable	http://localhost:8082/uploads/images/1747224122190_Rice.jpg
9	Sorghum	Loamy, sandy loam, or clayey loam soils	nitrogen, phosphorus, and potassium	20°C to 30°C	Kharif or Rabi crop	Not suitable	http://localhost:8082/uploads/images/1747224256741_sorghum.jpg
10	Watermelon	Sandy loam soil	Nitrogen, phosphorus, potassium	21°C and 35°C	Kharif season	June to September	http://localhost:8082/uploads/images/1747330979850_watermelon.jpg
11	Strawberries	loamy soil	nitrogen, phosphorus, potassium,	18°C and 25°C	Rabi crop	November to February	http://localhost:8082/uploads/images/1747361897868_strawberries.jpg
12	Corn	Loamy or sandy loam soils	Nitrogen, phosphorus, and potassium	21°C to 30°C	Kharif season	No snowing season	http://localhost:8082/uploads/images/1747676156696_corn.jpg

3. Seed Type Report:

Description: In this report admin can view the seed type details.

Seed Type Report																																											
Admin Panel Dashboard Approve Farmers Manage Crops Logout																																											
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4. Crop Stages Report:

Description: In this report admin can view the stages of crop.

Crop Stage Report																																																																																																																													
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5. Farmer Crop Schedule Report:

Description: In this report admin can view the farmer crop schedule details.

Admin Panel

- Dashboard
- Approve Farmers
- Manage Crops
- Logout

Farmer Crop Schedule
Back to Dashboard

Reports Table

farmer_id	crop_stage_id	start_date	end_date	activities	pesticides
21	1	2025-05-28	2025-05-28	Soil testing, land preparation, and seed selection.	Use pre-emergent herbicides like Pedimethalin to control weeds.
21	2	2025-05-29	2025-05-29	Sow seeds at the recommended depth and spacing.	No pesticides required at this stage.
21	3	2025-05-30	2025-06-07	Ensure adequate moisture for seeds germination. Monitor for pests.	Apply fungicides like Carbendazim if fungal diseases are observed.
21	4	2025-06-08	2025-06-27	Promote tiller growth with proper irrigation and nutrient management.	Use insecticides like Chlorpyrifos for pest control if needed.
21	5	2025-06-28	2025-07-27	Apply nitrogen fertilizers to boost growth. Monitor for pests and diseases.	Use herbicides like 2,4-D to manage broadleaf weeds.
21	6	2025-07-28	2025-08-26	Ensure proper irrigation. Monitor for pests and diseases.	Apply fungicides like Propiconazole for fungal infection.
21	7	2025-08-27	2025-09-15	Ensure even flowering by maintaining optimal conditions.	Use insecticides like Imidacloprid for aphid control if necessary.
21	8	2025-09-16	2025-10-05	Maintain irrigation and nutrient supply. Monitor for pests and diseases.	Apply fungicides like Tebuconazole to prevent fungal diseases.
21	9	2025-10-06	2025-10-25	Reduce irrigation to allow grains to mature.	No pesticides required at this stage.
21	10	2025-10-26	2025-11-04	Harvest the crop when grains are fully mature and moisture content is low.	No pesticides required at this stage.

Conclusion and Suggestions

7.1 Conclusion

In conclusion, the Application for Mazi Sheti – Farm Schedule System successfully aims to digitalize and streamline the daily operations of farmers, enhancing the efficiency and organization of farm activities. By allowing farmers to track, schedule, and receive reminders for crucial farming tasks such as planting, irrigation, fertilizing, and harvesting, the system improves time management and resource utilization, which ultimately leads to higher productivity and better farm outcomes.

The system helps farmers manage their time and resources better, leading to increased productivity and better results from their farming activities.

The system provides important tools like scheduling tasks, tracking activities, and giving reminders. It also includes detailed crop information to help farmers plan their activities more effectively.

The Application for Mazi Sheti – Farm Schedule System is a helpful tool for farmers to make their work easier and more organized. It improves farm management and can be made better with more features to meet the changing needs of farmers.

7.2 Future Enhancement

- Use AI to analyze crop images for early pest or disease detection, offering treatment suggestion.
- Connect farmer to real- time market prices and potential buyers for better decision-making.
- Offer virtual consultations with agricultural experts for advice.
- Build a social platform for farmers to share advice, experience, and farming tips.
- Provide offline mode for areas with limited internet access, syncing data when reconnected.
- Add a feature to connect farmers with local equipment rental services.
- Suggest the best crop to plant based on seasonal condition, and market demand.
- Allow farmer to connect with local government or NGOs for access to agriculture program.
- Provide recommendation for eco-friendly and sustainable pest control options.

7.3 Suggestions

- Use smart technology to recommend which crops to grow based on your land, weather, and market trends.
- Let farm owners assign tasks to workers, track their progress, and keep everyone organized.
- Connect the system to smart watering tools that automatically water crops based on soil moisture, saving water and improving crop health.
- Keep track of where your seeds come from and follow them all the way through to sale for better quality control and marketing.
- Track when to service your farming tools and machinery so they don't break down during important farming times.
- Add a marketplace where you can buy farming supplies or sell your produce directly to customers or buyers.
- Share a virtual tour of your farm to attract buyers or investors and show off your work.
- Provide feature which helps farmers share information with each other about crops, pests, or weather conditions to help everyone improve.

Annexure

- References

A) Book Reference –

Sr no	Book Name	Author	Edition	Publication	ISBN No
1	Microsoft SQL Server 2012 step by step	Patrick LeBlance	First Edition	Prentice Hall India Learning Private Limited	978-81-203-4763-2
2	Software Engineering	Pankaj Jalotes	First Edition	Wiley	978-81-265-2311-5

B) Web References –

<https://www.stackoverflow.com>

<https://www.W3School.com>

<https://start.spring.io>

<https://create-react-app.dev>

<https://www.mysql.com>